

DEEPAK SINGH

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New Delhi, India

SUMMARY

AI/ML Engineer with defense-grade predictive maintenance experience on T-72 battle tanks. Proven ability to translate complex operational problems into deployable ML solutions — evidenced by national hackathon podium finishes. Seeking roles where AI drives measurable operational impact in high-stakes environments.

EDUCATION

Guru Gobind Singh Indraprastha University

New Delhi, India

Bachelor of Technology in Artificial Intelligence and Data Science (CGPA: 8.62/10.0)

Aug 2024 – July 2028

TECHNICAL SKILLS

Languages : Python, SQL, C++

Machine Learning & AI : TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Data Science : Data Preprocessing, EDA, Feature Engineering, Model Evaluation, Time-Series Analysis

Backend & Tools : FastAPI, Docker, Git, GitHub, Jupyter Lab, Render, Vercel

Relevant Courses : ML, DSA, DBMS, OOP, Statistics, Linear Algebra, Data Mining, AI

EXPERIENCE

AI/ML Intern, Indian Army (505 Army Base Workshop)

June 2026 – Aug 2026

Under Lt. Col. Narender Pal Singh Mann | Delhi Cantonment, Delhi

- Developed a **predictive engine health monitoring system** for T-72 battle tanks using LSTM-based anomaly detection and XGBoost classification, achieving **92% accuracy** in early fault detection — shifting maintenance from reactive to proactive and reducing unplanned downtime by an estimated **30%**.
- Designed an AI-powered **maintenance assistant** with a FastAPI backend and rule-based expert system, delivering real-time diagnostics and step-by-step repair recommendations — deployed on edge devices for field use, reducing mean-time-to-repair (MTTR) by **25%**.
- Engineered an end-to-end time-series pipeline processing **50+ sensor parameters** (temperature, vibration, pressure) using Pandas and NumPy — applied ARIMA for forecasting, Isolation Forest for anomaly scoring, and SHAP for interpretable feature importance.
- Collaborated directly with defense engineers to translate operational pain points into ML-driven solutions — presented weekly dashboards and predictive insights to Lt. Col. Mann, influencing maintenance strategies across **100+ T-72 tanks**.

PROJECTS

Movie Recommendation System | Python, Scikit-learn, NLP, Pandas

Feb 2026 – Mar 2026

- Used **TMDB 5000** dataset containing movie metadata (overview, genres, cast, crew, keywords).
- Performed **text preprocessing** using NLTK — tokenization, stopword removal, and stemming for feature normalization.
- Applied **CountVectorizer** to convert text data into numerical feature vectors.
- Computed **cosine similarity** between movies to measure content-based similarity.
- Built a recommendation function that returns **top-5 similar movies** for any given title.
- Evaluated recommendations manually for relevance and diversity across genres.
- **Tech Stack**: Python, Pandas, NumPy, Scikit-learn, NLTK, Jupyter Lab

AWARDS & CERTIFICATIONS

- **2nd Position, Smart Delhi Ideation 2024** — Awarded by **Hon'ble Lt. Governor of Delhi, Shri Vinai Kumar Saxena** at Vigyan Bhawan. Ranked 2nd among 1000+ teams in a GGSIPU-organized innovation challenge.
- **2nd Runner-Up, Smart India Hackathon 2025** — Built an AI-powered Train Traffic Control system maximizing throughput using ML and data science.